

Vinod Vijay Nigade, PhD

ML Performance & Efficiency @ eBay Amsterdam

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Research Interests

I have around ten years of research and development experience in computer systems, including machine learning systems, systems for video analytics, data storage, and distributed systems. My current research focus is on developing *efficient [networked] systems* for machine learning (ML) [inference]. During my PhD, I worked on designing an inference serving system for latency-sensitive video analytics applications by leveraging the strengths of various modern computing platforms, spanning end devices, the edge, and the cloud.

Professional Experience

- 2025 – ■ **Machine Learning Performance Engineer**
eBay, Amsterdam, The Netherlands
Improving performance and efficiency of ML systems and models.
- 2024 – 2024 ■ **Guest Researcher**
Vrije Universiteit Amsterdam, The Netherlands
Worked in the HPDC group of the Computer Systems section.
- 2023 – 2024 ■ **Postdoctoral Researcher**
Vrije Universiteit Amsterdam, The Netherlands
Worked in the HPDC group of the Computer Systems section.
- 2017 – 2018 ■ **Principal Software Engineer**
Izel Technologies, Pune, India
Worked on a minimum viable product for source code analysis and code completion.
- 2017 – 2017 ■ **Software Engineer**
WEBB Traders, Amsterdam, The Netherlands
C++ developer for developing the proprietary trading system.
- 2015 – 2015 ■ **Research Assistant**
CNCR, Vrije Universiteit Amsterdam, The Netherlands
Wrote software to control the internally designed hardware to study rodent behavior.
- 2011 – 2014 ■ **Software Engineer**
Symantec Corporation (now Veritas), Pune, India
Worked on veritas volume manager (VxVM), specifically on veritas volume replicator.

Education

- 2018 – 2023 ■ **Ph.D. in Computer Science (Best ASCI Thesis Award 2023)**
Vrije Universiteit Amsterdam, Amsterdam, The Netherlands
Thesis title: *Latency-Critical Inference Serving for Deep Learning*
Advisors: prof. Henri Bal and prof. Lin Wang
- 2014 – 2016 ■ **M.Sc. cum laude in Parallel and Distributed Computer System**
Vrije Universiteit Amsterdam, Amsterdam, The Netherlands
Thesis title: *Efficiently Detecting Use-After-Free Exploits in Multi-Threaded Applications*
Advisors: dr. Cristiano Giuffrida and dr. Erik van der Kouwe
- 2007 – 2011 ■ **B.Tech. in Computer Science and Engineering**
Walchand College of Engineering, Sangli, India
Thesis title: *Enhanced Implementation of Evolutionary Optimization Service Times in IVRS*
Advisors: dr. B. F. Momin

Publications & Patents

Conference Proceedings

- 1 G. H. Apostolo, P. Bauszat, **V. Nigade**, H. E. Bal, and L. Wang, "Uirapuru: Timely video analytics for high-resolution steerable cameras on edge devices," in *ACM Conference On Mobile Computing And Networking (ACM MobiCom)*, 2025.
- 2 G. Liu*, **V. Nigade***, H. Bal, and L. Wang, "A little certainty is all we need: Discovery and synchronization acceleration in battery-free iot," in *ACM Asia-Pacific Workshop on Networking (APNET)*, 2024.
- 3 K. Razavi, S. D. Fard, G. Karlos, **V. Nigade**, M. Mühlhäuser, and L. Wang, "NetNN: Neural intrusion detection system in programmable networks," in *IEEE Symposium on Computers and Communications (ISCC)*, 2024. 🏆 **Best Paper Award (second place)**.
- 4 G. H. Apostolo, P. Bauszat, **V. Nigade**, H. E. Bal, and L. Wang, "Live video analytics as a service," in *ACM European Workshop on Machine Learning and Systems (ACM EuroSys EuroMLSys)*, 2022, pp. 37–44.
- 5 **V. Nigade**, P. Bauszat, H. Bal, and L. Wang, "Jellyfish: Timely inference serving for dynamic edge networks," in *IEEE Real-Time Systems Symposium (RTSS)*, 2022, pp. 277–290. 🏆 **Outstanding Paper Award**.
- 6 K. Razavi, G. Karlos, **V. Nigade**, M. Mühlhäuser, and L. Wang, "Distributed DNN serving in the network data plane," in *International Workshop on P4 in Europe (EuroP4)*, 2022, pp. 67–70.
- 7 **V. Nigade***, R. Winder*, H. Bal, and L. Wang, "Better never than late: Timely edge video analytics over the air," in *ACM Conference on Embedded Networked Sensor Systems (ACM SenSys AIChallengeIoT)*, 2021, pp. 426–432.
- 8 **V. Nigade**, L. Wang, and H. Bal, "Clownfish: Edge and cloud symbiosis for video stream analytics," in *IEEE/ACM Symposium on Edge Computing (SEC)*, 2020, pp. 55–69.
- 9 E. Van Der Kouwe, **V. Nigade**, and C. Giuffrida, "Dangsan: Scalable use-after-free detection," in *ACM European Conference on Computer Systems (EuroSys)*, 2017, pp. 405–419.

Journal Articles

- 1 **V. Nigade**, P. Bauszat, H. Bal, and L. Wang, "Inference serving with end-to-end latency slos over dynamic edge networks," *Real-Time Systems*, pp. 1–52, 2024.
- 2 R. De Haan, J. Lim, S. A. Van der Burg, A. W. Pieneman, **V. Nigade**, H. D. Mansvelder, and C. P. De Kock, "Neural representation of motor output, context and behavioral adaptation in rat medial prefrontal cortex during learned behavior," *Frontiers in Neural Circuits*, vol. 12, p. 75, 2018.

Patents

- 1 **V. Nigade** and M. Soundalgekar, *Distributed replication in cluster environments*, US Patent 9,600,553, 2017.

Awards & Grants

May. 2024	🏆 Best ASCI PhD Thesis Award 2023 Awarded at CompSys Dutch Conference 2024 for outstanding ASCI-aligned PhD research.
Dec. 2022	🏆 Outstanding Paper Award, IEEE RTSS
Mar. 2022	🏆 NVIDIA Academic Hardware Grant Two A30 tensor core GPUs.
2014 – 2016	🏆 VUFP Scholarship, Vrije Universiteit Amsterdam, The Netherlands A full merit scholarship for strongly motivated students.
2011 – 2014	🏆 Seven employee awards at Symantec Corporation, Pune, India Including innovation awards, customer-centric awards, and the best study paper award.

Awards & Grants (continued)

2007 – 2011 📖 Four programming and project competition awards during B.Tech.

Open Source

- Jellyfish 📖 A Soft Real-Time Inference Serving System
🔗 <https://github.com/vuhpdc/jellyfish>
- Clownfish 📖 Edge and Cloud Symbiosis for Video Stream Analytics
🔗 <https://github.com/vuhpdc/clownfish>
- DangSan 📖 Efficiently Detecting Use-After-Free Exploits in Multi-Threaded Applications
🔗 <https://github.com/vnigade/DangSan>
- Whisker 📖 Software to control the operational characteristics of a behavioural apparatus
🔗 <https://github.com/vnigade/Whisker>

Academic Services

Technical Reviewer

- 2025 📖 IEEE Transactions on Services Computing
- Dec. 2023 📖 IEEE/ACM UCC INTEL4EC
- Apr. 2023 📖 Digital Enablers of the Computing Continuum Systems, NWO ICT Open
- Dec. 2022 📖 IEEE/ACM UCC INTEL4EC

Thesis Supervision

- Aug. 2022 📖 Rishikumar Radhakrishnan, M.Sc. Project, Vrije Universiteit Amsterdam

Teaching

- 2020 – 2022 📖 Teaching Assistant, Advanced Network Programming, Vrije Universiteit Amsterdam

Skills

- Programming & Scripting Languages 📖 C/C++, Python, Shell (Bash), Java (novice)
- Frameworks & Libraries 📖 Pytorch, Tensorflow, OpenCV, gRPC, Gstreamer
- Version Control Systems 📖 Git, CVS (a long time back)
- Editors & IDEs 📖 Vim, VS Code

References

Prof. Henri Bal

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Prof. Lin Wang

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Dr. Pablo Bauszat

Scientific Researcher, High Performance Distributed Computing
Department of Computer Science
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